

RISKGATE Technical Overview and Trade Model Specification

Prepared for the RiskGate local-first MVP

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Abstract

RISKGATE is a local-first portfolio and options risk-management application. It was built to manually track portfolio holdings, import GBM portfolio spreadsheets, keep separate account-level portfolios, monitor watchlist tickers, document option trade ideas, score those trade ideas with a rules-based model, and assign conservative risk sizes. This document explains what has been built, how the app functions, how data is stored, and the exact mathematical approach behind the current trade approval model.

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1 Purpose of the Application

RISKGate was designed as a decision-support system for manual stock and options trading. Its purpose is not to execute orders, predict the future, or connect directly to a broker account. Instead, it helps enforce process discipline:

- keep multiple portfolios separate;
- import manually exported GBM portfolio data;
- track positions, cash, exposure, and allocation;
- maintain a watchlist of followed tickers;
- force every options trade idea to include a written thesis, invalidation level, max loss, and exit plan;
- score trade ideas using market, technical, options-quality, and trade-quality inputs;
- reject incomplete or unsafe options setups;
- calculate a suggested risk amount using account-level caps; and
- record entered and closed trades in a journal for later review.

The key rule of the system is:

No options trade should be approved unless the written plan, weighted score, liquidity checks, expiration rules, and portfolio risk caps all pass.

2 Technology Stack

The current MVP is a local web application, intended to run on the user's machine. It is not yet packaged as a Windows executable.

Layer	Implementation
Frontend	Next.js, React, TypeScript
Styling	Tailwind CSS
Database	SQLite
Database client	Prisma Client
Charts	Recharts
Icons	Lucide React
Excel import	<code>read-excel-file</code> package
Quote refresh	Yahoo Finance public chart endpoint, used as an unofficial read-only source
Runtime	Node.js local development server

2.1 Local-first Design

The local-first design was chosen because this app handles financial information. The MVP avoids:

- storing GBM or broker passwords;
- scraping broker login pages;
- syncing private data to the cloud;
- automated trading;
- order placement APIs.

This makes the first version safer. The app can later be wrapped with Tauri or Electron to create a Windows executable, but the current implementation is optimized for fast iteration.

3 Application Modules and Screens

3.1 Sidebar and Portfolio Switching

The sidebar contains navigation and a portfolio selector. The current app supports multiple portfolios. A selected portfolio is carried through the app using the URL query parameter:

```
?portfolio=<portfolioId>
```

This matters because the app does not treat all data as one combined account. Positions, watchlist items, trade ideas, journal stats, and risk settings are scoped to the selected portfolio.

3.2 Dashboard

The Dashboard summarizes the selected portfolio.

It shows:

- total tracked equity;
- cash;
- stock and ETF exposure;
- options exposure;
- options risk used;
- monthly options loss cap;
- open trade count;
- model-approved trade ideas;
- allocation chart;
- equity curve;
- risk exposure chart;
- recent trade ideas and their decisions.

3.3 Portfolio

The Portfolio page lets the user manually create, edit, and delete positions. It also contains the GBM Excel import panel.

Each position has:

- symbol;
- asset type: stock, ETF, option, or cash;
- market: MX or US;
- quantity;
- average cost;
- current price;
- currency;
- account/source label;
- notes.

3.4 GBM Excel Import

The Portfolio page includes an *Import GBM Excel* form. It accepts a GBM portfolio detail workbook. The first supported format is the GBM USA portfolio spreadsheet with rows like:

Emisora/Fondo	Titulos	Costo promedio	Precio mercado	...	Valor mercado	...
	META					
	NVDA					
	QQQ					
	SPY					
	efectivo					

The importer:

1. reads the uploaded Excel file;
2. ignores section headers such as **Mercado de Capitales USA**, **Liquidez**, and repeated table headers;
3. parses ticker rows;
4. converts **efectivo** into a **CASH** position;
5. identifies common ETFs such as **SPY**, **QQQ**, **V00**, etc.;
6. optionally converts USD values to MXN using the user-entered USD/MXN exchange rate;
7. either replaces all positions in the selected portfolio or updates matching symbols and inserts new ones.

3.5 Watchlist

The Watchlist page stores followed tickers for each portfolio. The app currently seeds common market and holding symbols:

SPY, QQQ, ^VIX, META, NVDA, AAPL

The user can:

- add tickers;
- edit ticker metadata;
- delete tickers;
- manually edit latest price values;
- refresh a single quote;
- refresh all Yahoo-sourced quotes in the selected portfolio.

The watchlist currently stores:

- symbol;
- name;
- market;
- currency;
- data source;
- current price;
- previous close;
- daily change;
- daily change percentage;
- last updated timestamp;

- notes.

At this stage, watchlist quote data is not yet used to automatically calculate RSI, moving averages, market breadth, or options-chain liquidity. It is the foundation for that future automation.

3.6 Trade Ideas

The Trade Ideas page is the core risk gate. Every possible options trade should be written here before entry.

The trade idea form captures:

- symbol;
- direction: bullish, bearish, or neutral;
- strategy: long call, long put, debit spread, protective put;
- thesis;
- catalyst;
- invalidation level;
- target price;
- expiration date;
- premium cost;
- max loss;
- expected reward;
- exit plan;
- market-regime inputs;
- technical inputs;
- options-quality inputs;
- chasing flag.

The app evaluates and stores:

- market regime score;
- technical score;
- options quality score;
- trade quality score;
- total score;
- decision;
- suggested risk in MXN;
- reasons;
- warnings.

3.7 Risk Model

The Risk Model page shows the weighted model, decision bands, sizing formula, risk usage, scored ideas, and recent model results.

3.8 Model Guide

The Model Guide explains the logic behind the model in the app. It includes:

- the weighted-score structure;

- point adjustments for every module;
- hard stops;
- sizing formula;
- watchlist tracking;
- limitations;
- future data integrations.

3.9 Journal

The Journal records entered and closed trades. It tracks:

- linked trade idea;
- entry date;
- exit date;
- entry price;
- exit price;
- realized P&L;
- result: win, loss, breakeven;
- mistake tags;
- notes.

The app calculates:

- win rate;
- average win;
- average loss;
- expectancy;
- total options P&L.

3.10 Settings

The Settings page manages:

- portfolio accounts;
- account names and institutions;
- account base currencies;
- account notes;
- total portfolio value;
- options sleeve;
- base risk per trade;
- monthly options loss cap;
- open options risk cap.

4 Database Model

The SQLite database is accessed through Prisma. The major entities are:

Model	Purpose
-------	---------

Portfolio	Represents a separate account or portfolio, such as one GBM account per person.
Position	Stores manual or imported holdings, scoped to a portfolio.
WatchlistItem	Stores followed tickers and quote data, scoped to a portfolio.
TradeIdea	Stores pre-trade option ideas, inputs, scores, decision, warnings, and suggested risk.
ModelScore	Stores a normalized score record linked one-to-one to a trade idea.
RiskSettings	Stores portfolio-level risk limits and sizing parameters.
TradeJournal	Stores entered and closed trades linked to trade ideas.

4.1 Portfolio Isolation

The following models carry a `portfolioId`:

- `Position`;
- `TradeIdea`;
- `RiskSettings`;
- `WatchlistItem`.

This means each account can have its own holdings, watchlist, trade ideas, and risk settings. Journal entries are linked through trade ideas, so they are indirectly portfolio-scoped.

5 Portfolio and Journal Calculations

5.1 Position Calculations

For a position i :

$$\text{MarketValue}_i = \text{Quantity}_i \times \text{CurrentPrice}_i, \quad (1)$$

$$\text{CostBasis}_i = \text{Quantity}_i \times \text{AverageCost}_i, \quad (2)$$

$$\text{UnrealizedPnL}_i = \text{MarketValue}_i - \text{CostBasis}_i, \quad (3)$$

$$\text{UnrealizedPnLPercent}_i = \begin{cases} \frac{\text{UnrealizedPnL}_i}{\text{CostBasis}_i} \times 100, & \text{if } \text{CostBasis}_i > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

Portfolio weight:

$$\text{Weight}_i = \frac{\text{MarketValue}_i}{\text{TotalPortfolioValue}} \times 100. \quad (5)$$

5.2 Exposure Calculations

The app groups position values by asset type:

$$\text{Cash} = \sum_{i \in \text{cash}} \text{MarketValue}_i, \quad (6)$$

$$\text{StocksAndETFs} = \sum_{i \in \{\text{stock}, \text{etf}\}} \text{MarketValue}_i, \quad (7)$$

$$\text{OptionsExposure} = \sum_{i \in \text{option}} \text{MarketValue}_i. \quad (8)$$

5.3 Journal Statistics

For closed journal entries:

$$\text{WinRate} = \frac{\# \text{WinningTrades}}{\# \text{ClosedTrades}} \times 100, \quad (9)$$

$$\text{AverageWin} = \frac{\sum \text{PositiveRealizedPnL}}{\# \text{WinningTrades}}, \quad (10)$$

$$\text{AverageLoss} = \frac{\sum \text{NegativeRealizedPnL}}{\# \text{LosingTrades}}, \quad (11)$$

$$\text{Expectancy} = \left(\frac{\text{WinRate}}{100} \right) \text{AverageWin} + \left(\frac{\text{LossRate}}{100} \right) \text{AverageLoss}. \quad (12)$$

Average loss is stored as a negative number in the current implementation, so expectancy naturally subtracts losses.

6 Trade Idea Model: Mathematical Specification

6.1 High-level Structure

The model has four scoring modules:

Module	Weight	Purpose
Market regime	30%	Broad market context using SPY, QQQ, VIX, breadth, and macro risk.
Technical setup	30%	Directional alignment with trend, moving averages, RSI, volume, support/resistance.
Options quality	25%	Contract quality using spread, volume, open interest, DTE, IV rank, and premium size.
Trade quality	15%	Discipline of the written plan, catalyst, invalidation, exit, reward/risk, and chasing flag.

Every module follows the same pattern:

$$\text{ModuleScore} = \text{clamp}(\text{BaseScore} + \sum \text{RuleAdjustments}, 0, 100). \quad (13)$$

The clamp function is:

$$\text{clamp}(x, 0, 100) = \min(100, \max(0, x)). \quad (14)$$

The total score is:

$$\text{TotalScore} = 0.30M + 0.30T + 0.25O + 0.15Q, \quad (15)$$

where:

- M is market regime score;
- T is technical setup score;
- O is options quality score;
- Q is trade quality score.

6.2 Market Regime Score

The market score starts at 50:

$$M_0 = 50. \quad (16)$$

The current implementation is:

$$\begin{aligned} M = & \text{clamp}(50 + 15\mathbb{I}\{\text{SPY bullish}\} - 15\mathbb{I}\{\text{SPY bearish}\} \\ & + 15\mathbb{I}\{\text{QQQ bullish}\} - 15\mathbb{I}\{\text{QQQ bearish}\} \\ & + 10\mathbb{I}\{\text{breadth strong}\} - 10\mathbb{I}\{\text{breadth weak}\} \\ & + 5\mathbb{I}\{\text{VIX low}\} - 10\mathbb{I}\{\text{VIX elevated}\} - 25\mathbb{I}\{\text{VIX spiking}\} \\ & + 5\mathbb{I}\{\text{macro risk low}\} - 20\mathbb{I}\{\text{macro risk high}\}, 0, 100). \end{aligned} \quad (17)$$

6.2.1 Interpretation

A bullish trade gets a better environment when SPY and QQQ are bullish, breadth is strong, VIX is low, and macro risk is low. A high-volatility or high-macro-risk environment penalizes the score because options can become expensive and direction can become less reliable.

6.3 Technical Setup Score

The technical score starts at 50:

$$T_0 = 50. \quad (18)$$

The current implementation is:

$$\begin{aligned} T = & \text{clamp}(50 + 18\mathbb{I}\{\text{trend up and direction bullish}\} + 18\mathbb{I}\{\text{trend down and direction bearish}\} \\ & + 12\mathbb{I}\{\text{trend sideways and direction neutral}\} - 15\mathbb{I}\{\text{trend up and direction bearish}\} \\ & - 15\mathbb{I}\{\text{trend down and direction bullish}\} \\ & + 8\mathbb{I}\{\text{price above 20MA and not bearish}\} - 6\mathbb{I}\{\text{price above 20MA and bearish}\} \\ & + 10\mathbb{I}\{\text{price above 50MA and not bearish}\} - 8\mathbb{I}\{\text{price above 50MA and bearish}\} \\ & + 8\mathbb{I}\{\text{RSI oversold and bullish}\} + 8\mathbb{I}\{\text{RSI overbought and bearish}\} \\ & - 8\mathbb{I}\{\text{RSI overbought and bullish}\} \\ & + 8\mathbb{I}\{\text{volume strong}\} - 8\mathbb{I}\{\text{volume weak}\} \\ & + 10\mathbb{I}\{\text{support/resistance good}\} - 15\mathbb{I}\{\text{support/resistance poor}\}, 0, 100). \end{aligned} \quad (19)$$

6.3.1 Interpretation

The technical module rewards alignment. Bullish ideas are rewarded when the asset is in an uptrend and above moving averages. Bearish ideas are rewarded when the asset is in a downtrend. The score penalizes fighting the trend, weak volume, and poor support/resistance structure.

6.4 Options Quality Score

The options quality score starts at 65:

$$O_0 = 65. \quad (20)$$

The higher base reflects that the model is not trying to reward a subjective thesis here; it is checking whether the contract is tradable. The current formula is:

$$\begin{aligned} O = \text{clamp}(65 + 15\mathbb{I}\{\text{spread} \leq 5\% \} - 8\mathbb{I}\{\text{spread} > 5\% \} \\ - 18\mathbb{I}\{\text{spread} > 10\% \} - 35\mathbb{I}\{\text{spread} > 15\% \} \\ + 8\mathbb{I}\{\text{option volume} \geq 500 \} - 15\mathbb{I}\{\text{option volume} < 50 \} \\ + 8\mathbb{I}\{\text{open interest} \geq 1000 \} - 15\mathbb{I}\{\text{open interest} < 100 \} \\ + 10\mathbb{I}\{21 \leq \text{DTE} \leq 60 \} - 40\mathbb{I}\{\text{DTE} < 7 \} \\ - 8\mathbb{I}\{\text{DTE} > 120 \} + 5\mathbb{I}\{\text{premium} \leq 0.5\% \text{ of portfolio} \} \\ - 30\mathbb{I}\{\text{premium} > 1\% \text{ of portfolio} \} - 15\mathbb{I}\{\text{IV rank} > 80 \}, 0, 100). \end{aligned} \quad (21)$$

6.4.1 Interpretation

This module is intentionally strict. A good stock setup can still be a bad options trade if:

- the bid/ask spread is too wide;
- volume is too low;
- open interest is too low;
- expiration is too close;
- premium is too large relative to the portfolio;
- implied volatility is unusually high.

6.5 Trade Quality Score

The trade quality score starts at 40:

$$Q_0 = 40. \quad (22)$$

Let:

$$R = \begin{cases} \frac{\text{ExpectedReward}}{\text{MaxLoss}}, & \text{if ExpectedReward exists and MaxLoss} > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (23)$$

Then:

$$\begin{aligned}
Q = & \text{clamp}(40 + 20\mathbb{I}\{\text{thesis length} \geq 30\} + 8\mathbb{I}\{\text{catalyst exists}\} \\
& + 12\mathbb{I}\{\text{invalidation level exists}\} + 12\mathbb{I}\{\text{exit plan exists}\} \\
& + 10\mathbb{I}\{R \geq 2\} - 12\mathbb{I}\{0 < R < 1.5\} - 25\mathbb{I}\{\text{is chasing}\}, 0, 100).
\end{aligned} \tag{24}$$

6.5.1 Interpretation

The trade quality score rewards disciplined planning. It is less about market forecasting and more about whether the trader has written down:

- why the trade exists;
- what invalidates the setup;
- how much can be lost;
- when to exit;
- whether the reward justifies the risk.

7 Decision Logic

7.1 Initial Decision

After calculating total score:

Total Score	Initial Decision
$S < 60$	reject
$60 \leq S < 75$	watchlist
$75 \leq S < 85$	approved small
$S \geq 85$	approved normal

7.2 Hard Stop Override

Even if the total score is high, hard stops force the final decision to reject.

Hard stops are:

- no written thesis;
- no invalidation level;
- max loss missing or ≤ 0 ;
- no exit plan;
- bid/ask spread greater than 15%;
- days to expiration less than 7;
- premium greater than 1% of portfolio;
- option volume less than 25 and open interest less than 100.

In symbols:

$$\text{FinalDecision} = \begin{cases} \text{reject}, & \text{if any hard stop is true,} \\ \text{InitialDecision}, & \text{otherwise.} \end{cases} \tag{25}$$

8 Risk Sizing Model

8.1 Base Settings

For the default 100,000 MXN account:

Setting	Default
Total portfolio value	100,000 MXN
Options sleeve percent	3%
Max options sleeve	3,000 MXN
Base risk per trade percent	0.5%
Base risk per trade	500 MXN
Max monthly options loss percent	1%
Max monthly options loss	1000 MXN
Max open options risk percent	2%
Max open options risk	2,000 MXN

8.2 Uncapped Suggested Risk

The uncapped risk formula is:

$$\text{UncappedRisk} = B \times C(S) \times L(\text{spread}) \times G(\text{regime, direction}) \times V(\text{IV rank}), \quad (26)$$

where:

- B is base risk per trade in MXN;
- $C(S)$ is confidence multiplier;
- L is liquidity multiplier;
- G is market regime multiplier;
- V is volatility multiplier.

8.3 Confidence Multiplier

$$C(S) = \begin{cases} 0, & S < 75, \\ 0.25, & 75 \leq S < 80, \\ 0.50, & 80 \leq S < 85, \\ 0.75, & 85 \leq S < 90, \\ 1.00, & S \geq 90. \end{cases}$$

8.4 Liquidity Multiplier

$$L(\text{spread}) = \begin{cases} 0, & \text{spread} > 15\%, \\ 0.25, & 10\% < \text{spread} \leq 15\%, \\ 0.50, & 5\% < \text{spread} \leq 10\%, \\ 1.00, & \text{spread} \leq 5\%. \end{cases}$$

8.5 Market Regime Multiplier

The current code uses SPY trend as the primary regime proxy:

$$G = \begin{cases} 1.00, & \text{SPY bullish and trade bullish,} \\ 1.00, & \text{SPY bearish and trade bearish,} \\ 0.50, & \text{SPY neutral,} \\ 0.25, & \text{otherwise.} \end{cases}$$

8.6 Volatility Multiplier

$$V(\text{IV rank}) = \begin{cases} 1.00, & \text{IV rank missing,} \\ 0.25, & \text{IV rank} > 80, \\ 0.50, & 60 < \text{IV rank} \leq 80, \\ 0.75, & \text{IV rank} < 20, \\ 1.00, & 20 \leq \text{IV rank} \leq 60. \end{cases}$$

8.7 Final Caps

The final risk is capped by portfolio-level remaining limits:

$$\text{RemainingSleeve} = \max(\text{MaxOptionsSleeve} - \text{OptionsRiskUsed}, 0), \quad (27)$$

$$\text{RemainingMonthlyLoss} = \max(\text{MonthlyLossCap} - \text{MonthlyLossUsed}, 0), \quad (28)$$

$$\text{RemainingOpenRisk} = \max(\text{OpenRiskCap} - \text{OpenRiskUsed}, 0). \quad (29)$$

Then:

$$\text{SuggestedRisk} = \begin{cases} \min(\text{UncappedRisk}, \text{RemainingSleeve}, \text{RemainingMonthlyLoss}, \text{RemainingOpenRisk}), & \text{if applicable} \\ 0, & \text{otherwise} \end{cases} \quad (30)$$

9 Example Risk Calculation

Suppose:

- total score is 82;
- base risk is 500 MXN;
- bid/ask spread is 7%;
- SPY is bullish and the trade is bullish;
- IV rank is 42;
- all caps have enough room.

Then:

$$C(82) = 0.50, \quad L(7\%) = 0.50, \quad G = 1.00, \quad V(42) = 1.00.$$

$$\text{UncappedRisk} = 500 \times 0.50 \times 0.50 \times 1.00 \times 1.00 = 125 \text{ MXN.}$$

Rounded:

$$\text{SuggestedRisk} \approx 125 \text{ MXN.}$$

If any cap has less than 125 MXN remaining, the final suggested risk is reduced to that cap.

10 Current Limitations

- Market regime inputs are manually selected.
- Technical inputs are manually selected.
- Watchlist quotes do not yet calculate moving averages, RSI, or trend.
- Options chains are not yet fetched automatically.
- Bid, ask, mid, Greeks, implied volatility, volume, and open interest are entered manually.
- Strategy-specific payoff diagrams are not yet implemented.
- The model does not backtest rules against historical trades.
- The model does not calculate probability of profit.
- The model does not automatically detect earnings dates or macro events.
- The model does not yet adjust for sector concentration or correlation.

11 Future Robustness Layers

11.1 Layer 1: Read-only Market Data

The next logical step is a read-only Options Chain Analyzer. It would pull:

- expirations;
- strikes;
- bid and ask;
- midpoint;
- bid/ask spread percentage;
- option volume;
- open interest;
- implied volatility;
- Greeks, if available;
- days to expiration.

Then it could pre-fill the options-quality section of the trade idea form.

11.2 Layer 2: Technical Indicator Automation

The Watchlist can be extended to store historical prices. From that, RiskGate could calculate:

$$\text{SMA}_{20}(t) = \frac{1}{20} \sum_{k=0}^{19} P_{t-k}, \quad (31)$$

$$\text{SMA}_{50}(t) = \frac{1}{50} \sum_{k=0}^{49} P_{t-k}. \quad (32)$$

RSI could be calculated with the standard 14-period approach:

$$RS = \frac{\text{AverageGain}_{14}}{\text{AverageLoss}_{14}}, \quad (33)$$

$$RSI = 100 - \frac{100}{1 + RS}. \quad (34)$$

These indicators could automatically fill:

- trend;
- price versus 20MA;
- price versus 50MA;
- RSI condition;
- volume condition.

11.3 Layer 3: Strategy-specific Logic

Future versions should score strategies differently:

- long calls should require directional momentum and controlled premium;
- long puts should require downside trend confirmation or hedge rationale;
- debit spreads should calculate width, max gain, max loss, break-even, and reward/risk;
- protective puts should compare hedge cost against protected notional value and drawdown risk.

11.4 Layer 4: Backtesting and Feedback

The journal can be used to determine whether the model works:

- win rate by score band;
- average P&L by score band;
- expectancy by strategy;
- expectancy by module score;
- common mistake tags in losing trades;
- comparison of approved trades versus watchlist/rejected trades.

11.5 Layer 5: External Integrations

Potential data/integration providers:

- Tradier: <https://docs.tradier.com/reference/brokerage-api-markets-get-options-chains>
- Polygon.io: <https://polygon.io/docs/rest/options/overview>
- MarketData.app: <https://www.marketdata.app/docs/api/>
- Alpaca: <https://docs.alpaca.markets/docs/options-trading>

The recommended path is read-only data first. Trading automation should remain out of scope until the app has reliable data handling, testing, audit logging, and explicit confirmation flows.

12 How to Run the App

From the project folder:

```
npm install
npm run prisma:generate
npm run db:init
npm run seed
npm run db:migrate:watchlist
npm run dev
```

Then open:

```
http://127.0.0.1:3000
```

13 Files of Interest

File	Purpose
lib/risk.ts	Core scoring, decision, hard-stop, and sizing model.
lib/calculations.ts	Portfolio, P&L, risk usage, and journal statistics.
app/actions.ts	Server actions for CRUD, Excel import, scoring, settings, watchlist refresh.
prisma/schema.prisma	Database schema.
app/page.tsx	Dashboard.
app/portfolio/page.tsx	Portfolio positions and GBM Excel import.
app/watchlist/page.tsx	Ticker tracking and quote refresh.
app/trade-ideas/page.tsx	Trade idea form and score display.
app/risk-model/page.tsx	Risk model dashboard.
app/model-guide/page.tsx	In-app explanation of the scoring model.
app/journal/page.tsx	Trade journal.
app/settings/page.tsx	Portfolio accounts and risk settings.

14 Safety Notes

- Options are risky and can expire worthless.
- RISKGATE is a decision-support tool, not financial advice.
- The app should not store broker passwords.
- Broker scraping should be avoided.
- Read-only integrations should come before any order-entry workflow.
- The current model is a rule-based checklist, not a predictive machine-learning model.

15 Conclusion

RISKGATE now includes the core building blocks for disciplined options decision-making: portfolio isolation, manual and imported positions, watchlist tracking, a formal trade idea gate, a transparent scoring model, dynamic sizing, and a trade journal. The most important future improvement is to automate objective inputs while keeping the final approval manual. A read-only options-chain

and market-data layer would make the model more robust without introducing the risks of broker credential storage or automated trading.